Plasticity in Deep Reinforcement Learning

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Master’s Project

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Abstract

Deep reinforcement learning has arisen to achieve remarkable success in domains such as game playing, robotics, recommendation systems, control, and large-scale decision making[Mnih et al., 2015, Silver et al., 2017]. However, despite these successes, modern deep reinforcement learning systems frequently suffer from a phenomenon known as plasticity loss.

Plasticity in general is the ability of a system to change how it perceives and processes the data and be able to adapt. In the context of human, Neuroplasticity is the brain's remarkable capacity to reorganize, grow, and rewire its neural connections throughout your lifetime. Rather than being a fixed organ, it continuously adapts to new experiences, learning processes, and environmental stimuli, or following an injury. [Puderbaugh and Emmady, 2023]. In the context of deep learning though, it refers to a neural network's ability to adapt its predictions when new information becomes available. As training progresses, networks often become increasingly difficult to modify, resulting in slower learning, reduced adaptability, and performance degradation in non-stationary environments[Lyle et al., 2023, Ash and Adams, 2020].

As we use DRL systems more and more by the day, we need to find the root causes preventing neural networks to benefit from plasticity while solving tasks in the reinforcement learning regime.

The objective of this project is to execute a literature study and investigate the mechanisms responsible for plasticity loss and understand how this phenomenon affects deep reinforcement learning systems. We would also reproduce the results based on the work [Lyle et al., 2023] and argue about the effect of geometrical landscape of functions and their effect on learning.

Contents

1 Motivation

Plasticity is becoming increasingly important as reinforcement learning systems are deployed in environments that continuously evolve.

Unlike traditional supervised learning, reinforcement learning operates under fundamentally non-stationary targets, especially when using value-based learning methods such as learning Q-values.

As the learning progresses during an RL task, policies and state values continuously change, data distributions evolve during training and target moves as learning objectives depend on previous updates, for example in case of temporal-difference learning methods.

As tasks become more complex, the ability of neural networks to remain adaptable becomes increasingly important. For example on longer horizon tasks and continouos action environments there is no other feasible solution than to use neural networks. In that case we need to make sure that this tool is a reliable function approximator to learn the dynamics of a changing environment.

Several recent works suggest that plasticity loss may be one of the major factors limiting the scalability of deep reinforcement learning. Even in the works purely done on supervised learning methods, we see that training the model on some part of the dataset at first and training it again on the second parts results in poor generalization[Ash and Adams, 2020]. Other works in the DRL have shown that for example as off-policy methods rely on using a replay buffer and use the data to train a model, will create the issue that the model overfits to early experiences in a way that harms future learning[Nikishin et al., 2022].

Understanding and preventing this phenomenon would lead to:

- More stable RL algorithms
- Faster adaptation
- Better continual learning
- Improved sample efficiency
- More robust autonomous agents

2 Fundamentals

The roots of this work can be traced to several related research areas.

2.1 Stability-Plasticity Dilemma

A long-standing challenge in neuroscience and machine learning is balancing stability and adaptability. Systems must preserve useful knowledge while remaining capable of learning new information.

Excessive stability leads to rigidity and inability to learn on new signals. Excessive plasticity leads to catasrophic forgetting.

2.2 Primacy Bias

[Nikishin et al., 2022] have already demonstrated that early experiences exert disproportionately large influence on neural network representations.

Networks become increasingly biased toward information learned early in training, reducing their ability to adapt later.

Also [Ash and Adams, 2020] showed that networks can become increasingly difficult to retrain after prolonged optimization.

Their observations provide evidence that optimization itself changes trainability.

2.3 Plasticity Loss

The most comprehensive recent study is: [Lyle et al., 2023]

“Understanding Plasticity Loss in Neural Networks”

This paper provides strong evidence that plasticity loss is fundamentally related to changes in the geometry of the loss landscape rather than simply dead neurons, feature collapse or other previously mentioned improvements on the matter.

In the next section we cover the methodology used to carry out the experiments.

3 Methodology

3.1 Optimizer Dynamics Under Sudden Changes in Gradient Structure

One of the first experiments reproduced in this project investigates the effect of abrupt changes in the learning objective on the optimization process. Following the methodology proposed by **?**, a neural network is first trained on a classification task with a fixed mapping between inputs and labels. After the network has successfully learned the task, the labels are randomly reassigned while keeping the input distribution unchanged.

This procedure creates a sudden change in the gradient structure encountered by the optimizer. While the network architecture and data remain identical, the gradients required to solve the new task differ substantially from those that were previously learned. Consequently, the optimizer must rapidly adapt to an entirely new learning signal.

The results reported by Lyle et al. suggest that adaptive optimizers such as Adam may struggle under these conditions. The abrupt change in gradient statistics can produce excessively large parameter updates, causing many hidden ReLU units to become inactive. Once a significant portion of the hidden units have died, the representational capacity of the network is reduced and subsequent learning becomes increasingly difficult. This phenomenon provides one possible explanation for the loss of plasticity observed in neural networks.

To verify these findings, the optimizer experiment was reproduced and analyzed in the present study. The resulting behavior of the network and optimizer is shown in Figure 1.

3.2 Loss Landscape Analysis

A central hypothesis investigated throughout this project is that plasticity loss is fundamentally related to changes in the geometry of the loss landscape. To study this hypothesis, two complementary analyses are performed: Hessian spectrum analysis and gradient covariance analysis.

The Hessian matrix describes the local curvature of the loss function with respect to the model parameters. Since modern neural networks may contain millions of parameters, explicitly constructing the Hessian matrix is computationally infeasible. Therefore, this work follows the methodology proposed by **?**, which estimates the Hessian eigenspectrum using Hessian-vector products combined with the Lanczos

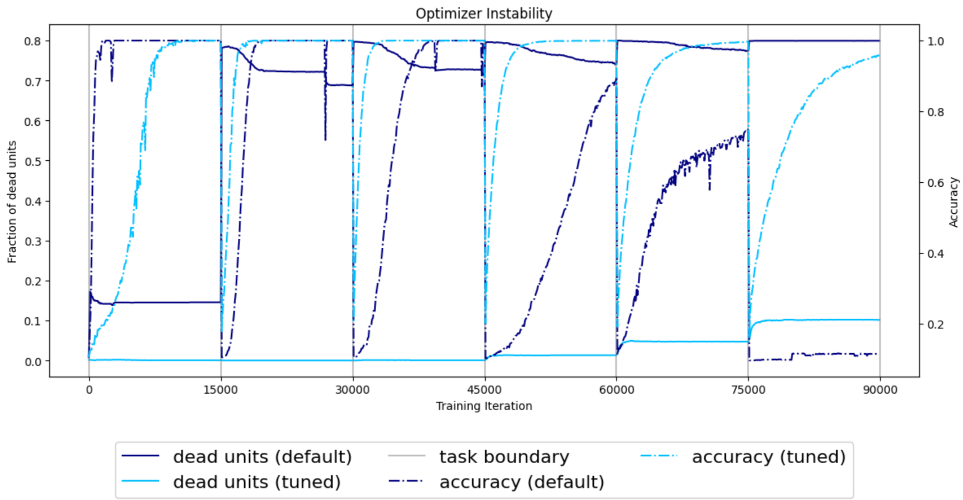

Figure 1: Effect of sudden changes in the learning objective on optimization dynamics and network plasticity.

algorithm.

The procedure begins by computing Hessian-vector products without explicitly forming the Hessian matrix. These products are then used within the Lanczos iteration to construct a low-dimensional tridiagonal approximation of the Hessian. The eigenvalues of the resulting matrix are subsequently smoothed using a Gaussian kernel density estimator to obtain an approximation of the Hessian eigenvalue density spectrum.

The Hessian spectrum provides information about several important geometric properties of the loss landscape, including:

- Landscape sharpness,
- Spectral outliers,
- Optimization conditioning,
- Curvature growth during training.

In addition to curvature analysis, gradient covariance matrices are computed to characterize the interaction between gradients generated by different samples. Positive covariance indicates cooperative learning directions, while negative covariance indicates interference between updates. The evolution of these covariance structures during training provides further insight into the mechanisms responsible for plasticity loss.

To isolate the network's ability to adapt to new information from the reinforcement learning objective itself, Hessian and gradient covariance measurements are computed using the probe loss introduced by **?**. The probe objective is defined as

$$L_{\text{probe}}(\theta) = \left(f_\theta(x)\text{sg}\left[f_\theta(x)\right] + \epsilon\right)^2, \tag{1}$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator and $\epsilon \sim \mathcal{N}(0, 1)$.

Unlike the temporal-difference loss used during DQN training, the probe loss measures how easily a trained network can adapt to randomly generated targets. Consequently, it serves as a direct measure of plasticity.

The results of the Hessian spectrum and gradient covariance analyses obtained in this study are shown in Figure 2.

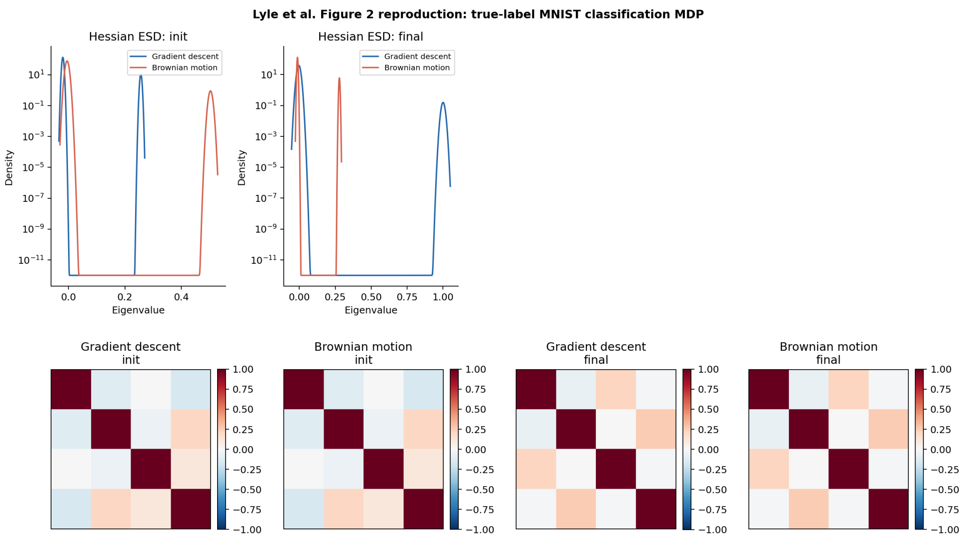

Figure 2: Loss landscape analysis using Hessian eigenspectra and gradient covariance matrices.

3.3 Falsification of Previously Proposed Mechanisms

Several explanations have previously been proposed to explain plasticity loss in reinforcement learning systems. Examples include feature rank collapse, weight norm growth, and the fraction of active ReLU units. While correlations between these quantities and plasticity have been observed in some experiments, correlation alone does not establish a causal relationship.

To investigate whether these mechanisms are causally responsible for plasticity loss, a falsification experiment similar to that presented by **?** is performed. A total of 128 DQN agents are trained across multiple Markov Decision Processes (MDPs) described in Section 2.2. During training, both the proposed explanatory variables and the measured plasticity are recorded.

The central idea of the experiment is that a genuine causal mechanism should exhibit a consistent relationship with plasticity across different environments. If a particular quantity truly causes plasticity loss, increases in that quantity should consistently correspond to decreases in plasticity regardless of the underlying task.

However, the experimental results reveal that this consistency does not hold. Across different environments, the relationship between the proposed explanatory variables and plasticity changes substantially. In some environments the correlation is positive, while in others it becomes negative. Consequently, no stable relationship can be established across all tasks.

These findings suggest that the previously proposed mechanisms are unlikely to constitute the fundamental cause of plasticity loss. Instead, they may represent secondary effects that emerge alongside the underlying phenomenon. This observation motivates the investigation of loss landscape geometry as a more fundamental explanation for the degradation of plasticity.

The results of the falsification experiments are shown in Figure 3.

Figure 3: Relationship between previously proposed explanatory variables and measured plasticity across multiple environments.

4 Results

5 Conclusion

This project investigated the mechanisms behind plasticity loss in neural networks, with a particular focus on deep reinforcement learning. The main objective was to understand why networks that have already undergone training often become less capable of adapting to new learning signals. To address this question, we studied the recent literature on plasticity loss, capacity loss, warm-starting, and value-function learning, and reproduced several central experiments from [**?**].

The experiments reproduced in this work support the view that plasticity loss is closely connected to the geometry of the loss landscape. In particular, the optimizer instability experiment showed that abrupt changes in the learning objective can significantly disturb the gradient structure and cause many hidden ReLU units to become inactive. This prevents the network from effectively using its representational capacity and can strongly reduce future learning ability. The Hessian and gradient covariance experiments further showed that training can move the network into regions of parameter space with sharper curvature and stronger gradient interference. These observations suggest that plasticity loss is not only a representational problem, but also an optimization problem.

Based on the reproduced experiments and the methods studied in lyle2023plasticity, the most effective interventions were those that made the loss curvature easier to optimize. In particular, methods such as layer normalization and categorical output representations were more successful than methods that only perturbed or regularized the parameters. This supports the hypothesis that preserving plasticity requires controlling the sharpness and conditioning of the loss landscape.

Layer normalization is especially relevant in this context because it changes the parameterization of the network in a way that can stabilize optimization. By normalizing intermediate activations, layer normalization can reduce sensitivity to scale changes and help prevent the network from entering extremely sharp regions of the loss landscape. In experiments of [**?**], the addition of layer normalization reduced plasticity loss in several architectures. This suggests that normalization methods may help preserve plasticity by producing smoother and better-conditioned optimization dynamics.

The results are also consistent with the conclusions of **?**, who argue that value functions in deep reinforcement learning should not necessarily be trained as standard regression problems. In conventional value-based RL, the network is trained to regress directly onto bootstrapped target values. This can be difficult because the targets are noisy, non-stationary, and unbounded. Regression objectives may therefore push the parameters into regions of the loss landscape from which later adaptation becomes difficult. In contrast, categorical methods such as two-hot encoding or HL-Gauss transform value learning into a classification-style problem. Instead of predicting a scalar value directly, the network predicts a distribution over a fixed support. This can make the learning problem better conditioned and more stable.

Further works can study methods to use to reduce the sharpness or smoothen the navigation of network in function space so that plasticity stays intact.

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